

Exterior of Airframe: Static wicks or static dischargers. Dischargers accumulate static electricity back to the surrounding air. Without them, it builds up & goes to other sharp extremities like the antennas, causes radio frequency

Cessna 172 Powerplant: Lycoming IO-360, 225 HP 4C

1. Openings on the nose to take in cool air, called Pits.
2. Alternator, vacuum pump to ~~keep~~ maintain gyroscopic instruments working, & heat produced for cockpit.
3. Intake, compression, combustion, & exhaust. Propeller directly connected to crankshaft which due to one cylinder always in combustion stroke this provides constant power, known as direct drive.

4. Carburetor & fuel injection system to deliver fuel to engine. This specific engine uses fuel injection with more advantage & modern. The carburetor relies mainly on mechanics, both system goal is to mix of air & fuel to the cylinder during intake stroke.

5. The engine is mounted on a metal frame, with some elastomeric mounts to ~~eliminate~~ minimize vibration, this assembly attached to main frame

6. Fuel system, has a fuel tank on each wing, which is somewhat above the cockpit & its length is about $\frac{1}{4}$ of the wing; On the other side of the spar carrythrough channel

7. To refuel, must connect grounding wire to at the exhaust or tie-down ring under the wing. It's unsafe for refueling

8. The fuel distributed from tank through fuel lines, can be connected to switch to draw fuel from left &/or right tank.

9. Small fuel reservoir to catch fuel from a return line from the engine. An electric auxiliary fuel pump is used to prime the engine during start up, then fuel goes through filter called gascolator & finally engine driven fuel pump during operation

10. Small vent under left wing to provide a pathway to stabilize the pressure inside the tank with the outside atmospheric pressure. This also helps with the flow of fuel.

Cessna 172 Control Surfaces:

1. Ailerons, rudder, & elevator. Secondary control are the flaps.
2. Ailerons controlled by a wires connected to the main control yoke, where the yokes are also connected, rudder
3. Elevator connected to horizontal stabilizer & elevator connected to vertical stabilizer.
4. The elevator uses the same mechanics of ailerons to be controlled from the yokes. This moves on its lateral axis. Ailerons move on longitudinal axis. There is an extra control called the trim tab, used to fine tune the elevator in different phase of flight & it counteracts the force the pilot has to use to constantly push or pull to maintain altitude. During takeoff, must be ensured or whatever is recommended.
5. Rudder controlled by pedals to go left or right. This happens in vertical axis which is the motion called yaw.



Controls by a vertical wheel from the inside of cockpit